Year 12 Further Maths - Spearman's rank correlation

Similar questions - Rankings and hypothesis tests

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Dev calculates the value of Spearman's rank correlation coefficient r_s for 8 pairs of rankings. Find the largest possible value of r_s that Dev can obtain that is less than 1. **[4]**

(b) A set of bivariate data consists of 6 pairs of values. It is known that for this data the value of Spearman's rank correlation coefficient is -1 but the value of Pearson's product-moment correlation coefficient is not -1. Sketch a possible scatter diagram illustrating the data. **[2]**

2 Nine runners took part in two races. The positions in which the runners finished in the two races are shown in the table.

Runner	A	B	C	D	E	F	G	H	I
First race	4	1	7	2	9	5	3	8	6
Second race	6	3	9	1	8	4	2	7	5

Test at the 5% significance level whether those runners who do better in one race tend to do better in the other. **[7]**

3 In a talent competition, two judges gave marks to a large number of contestants.

(a) The value of Spearman's rank correlation coefficient between the marks given to 8 randomly chosen contestants is $\frac{19}{21}$. Test, at the 1% significance level, whether there is evidence that the judges agree with each other. **[4]**

The marks given by the two judges to the 8 randomly chosen contestants were as follows, where x is a real number.

Contestant	A	B	C	D	E	F	G	H
Judge 1	54	58	61	63	67	70	72	78
Judge 2	52	49	58	60	56	64	x	71

(b) Use $r_s = \frac{19}{21}$ to determine the range of possible values of x. **[4]**

(c) Give a reason why it might be preferable to use the product-moment correlation coefficient rather than Spearman's coefficient in this context. **[1]**

Question 1

(a) $r_s = 1 - \dfrac{6\sum d^2}{n(n^2-1)}$. The smallest non-zero $\sum d^2$ occurs when two adjacent ranks are swapped: $d = \pm 1$ for two pairs, so $\sum d^2 = 2$ (it cannot be 1 as $\sum d$ must be 0). $r_s = 1 - \dfrac{6\times 2}{8\times 63} = 1 - \dfrac{12}{504} = \dfrac{41}{42} = 0.976$

(b) A scatter diagram showing six points with a monotonic decreasing but curved relationship (for example points on a decreasing curve such as $y = 1/x$), so the ranks are exactly reversed but the points do not lie on a straight line

Question 2

Ranks in order of first race (1 to 9) have second race positions $3,1,2,6,4,5,9,7,8$. d: $-2,1,1,-2,1,1,-2,1,1$, giving $d^2 = 4,1,1,4,1,1,4,1,1$ and $\sum d^2 = 18$. $r_s = 1 - \dfrac{6(18)}{9(80)} = 1 - \dfrac{108}{720} = 0.85$. H_0: no association; H_1: positive association. Critical value ($n = 9$, 5% one-tailed) $= 0.6000$. $0.85 > 0.6000$, so reject H_0: there is evidence that runners who do better in one race tend to do better in the other

Question 3

(a) H_0: no association between the judges' marks; H_1: there is association. Two-tailed 1% critical value for $n = 8$ is 0.8333. $\frac{19}{21} = 0.905 > 0.8333$ so reject H_0: there is evidence at the 1% level that the judges agree

(b) $r_s = \dfrac{19}{21}$ gives $1 - \dfrac{6\sum d^2}{504} = \dfrac{19}{21}$, so $\sum d^2 = 8$. Judge 1 ranks: A to H are $1,\ldots,8$. Judge 2's marks apart from x: $49(B) < 52(A) < 56(E) < 58(C) < 60(D) < 64(F) < 71(H)$. If G is ranked 7th ($64 < x < 71$): Judge 2 ranks $A = 2, B = 1, C = 4, D = 5, E = 3, F = 6, G = 7, H = 8$ and the values of d^2 are: A: $(1-2)^2 = 1$, B: $(2-1)^2 = 1$, C: $(3-4)^2 = 1$, D: $(4-5)^2 = 1$, E: $(5-3)^2 = 4$, F: $(6-6)^2 = 0$, G: 0, H: 0, total 8. Other positions of G give $\sum d^2 \neq 8$. So $64 < x < 71$

(c) The marks are interval data (not just ranks), and the product-moment coefficient uses the actual marks, so it uses more information (if the relationship is approximately linear and the data are bivariate normal)